1 Results: Baseline, Classical, and Quantum-Conditioned U-Net

This section summarizes the final three-way comparison among (i) the unconditioned U-Net baseline, (ii) the parameter-matched classical conditioned U-Net, and (iii) the variational quantum circuit (QNN) conditioned U-Net. The final configuration was selected on validation data only, frozen, and then evaluated on the locked test set. Results are aggregated over three seeds, $\{17, 42, 123\}$, on 2,000 test utterances with SNR values $\{-5, 0, 5, 10, 15, 20\}$ dB.

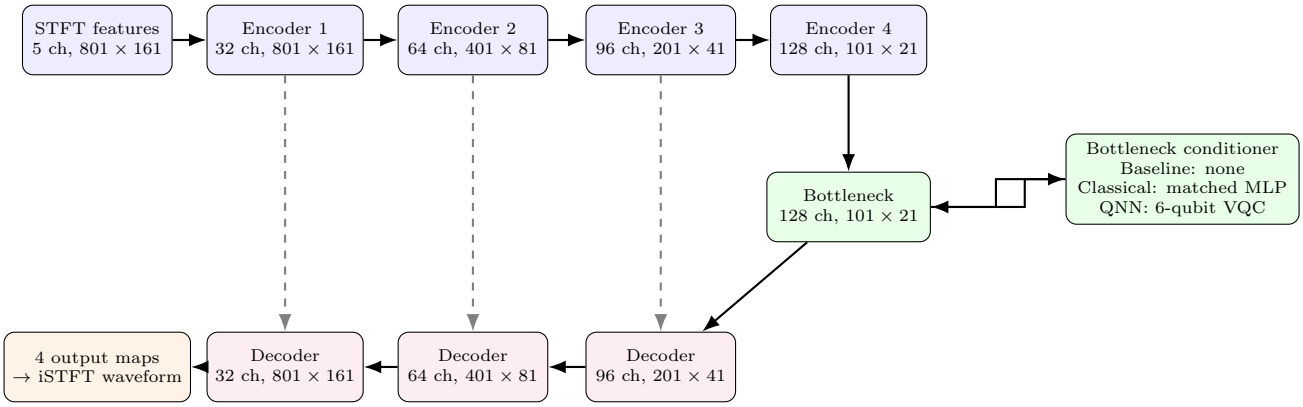
1.1 Common Backbone and Final Configuration

All three models share the same complex-spectral U-Net backbone. Audio is sampled at 16 kHz and cropped to 8 s segments. The short-time Fourier transform uses $N_{\mathrm{FFT}} = 320$, a 320-sample window, and a 160-sample hop. Five spectral input channels are constructed from the noisy STFT: normalized real part, normalized imaginary part, normalized log-magnitude, sine of phase, and cosine of phase. For an 8 s segment, these features have approximately 801×161 time–frequency locations. The encoder follows the standard U-Net contraction pattern: the spatial resolution decreases while channel depth increases. Three stride-2 convolutional downsampling operations produce approximately $801 \times 161 \to 401 \times 81 \to 201 \times 41 \to 101 \times 21$, with channels $32 \to 64 \to 96 \to 128$. The bottleneck operates at the deepest $128 \times 101 \times 21$ representation. The decoder then expands the resolution by bilinear interpolation to the corresponding skip sizes, $201 \times 41 \to 401 \times 81 \to 801 \times 161$, while reducing channels $96 \to 64 \to 32$. The output head predicts a magnitude mask, phase residual, real residual, and imaginary residual, which are matched to the original STFT resolution and converted back to waveform using iSTFT.

The main final settings are summarized in Table 1. The baseline is trained for 20 epochs. For the classical and QNN variants, the conditioner is first warmed up for 8 epochs and the complete network is then jointly fine-tuned for up to 35 epochs. The validation-selected core learning rates are 8×10^{-4} for the classical conditioner and 6.5×10^{-4} for the quantum conditioner.

Table 1: Main configuration used for the final three-seed comparison.

Setting	Value	Setting	Value
Sampling rate	16 kHz	Segment length	8 s
FFT / window / hop	320 / 320 / 160	Batch size	8
Encoder channels	32, 64, 96, 128	Dropout	0.08
Shared pretraining	20 epochs	Conditioner warmup	8 epochs
Joint fine-tuning	35 epochs	Seeds	17, 42, 123
Backbone LR	1×10^{-4}	Projection LR	2×10^{-4}
Classical core LR	8×10^{-4}	QNN core LR	6.5×10^{-4}
Weight decay	1×10^{-4}	Gradient clip	5.0



Encoder transitions use stride-2 convolution; decoder stages use bilinear interpolation before skip fusion.
Contraction: lower resolution / more channels → bottleneck → expansion: higher resolution / fewer channels.

Figure 1: U-Net architecture used in the final experiment. The encoder progressively reduces the time–frequency resolution while increasing channel depth; the decoder restores the resolution while fusing the three encoder skip features used by the implementation. Global conditioning is applied only at the bottleneck in the selected configuration.

1.2 Conditioning Input and Quantum Encoding

The baseline contains no conditioner. For both the classical and QNN variants, conditioning is inserted only at the deepest bottleneck. Importantly, the quantum model does *not* convert the signal into binary bits. Instead, continuous U-Net feature statistics are compressed into six real-valued angles.

The four encoder outputs have 32, 64, 96, and 128 channels and the processed bottleneck has 128 channels. For each tensor, the mean and standard deviation across the time–frequency dimensions are computed, giving

$$2(32 + 64 + 96 + 128 + 128) = 896 \tag{1}$$

scalar statistics. These are projected and normalized,

$$\mathbf{u} = \mathrm{LN}(W_{\mathrm{in}}\mathbf{s} + \mathbf{b}_{\mathrm{in}}), \tag{2}$$

followed by

$$\boldsymbol{\phi} = \pi \tanh(\mathbf{u}), \qquad \boldsymbol{\phi} \in [-\pi, \pi]^6. \tag{3}$$

Thus, both the classical and quantum conditioners receive exactly the same six-dimensional continuous representation. The distinction lies only in the 72-parameter core that maps these six values to a 12-dimensional latent representation. Figure 2 summarizes the quantum encoding and its matched classical control.

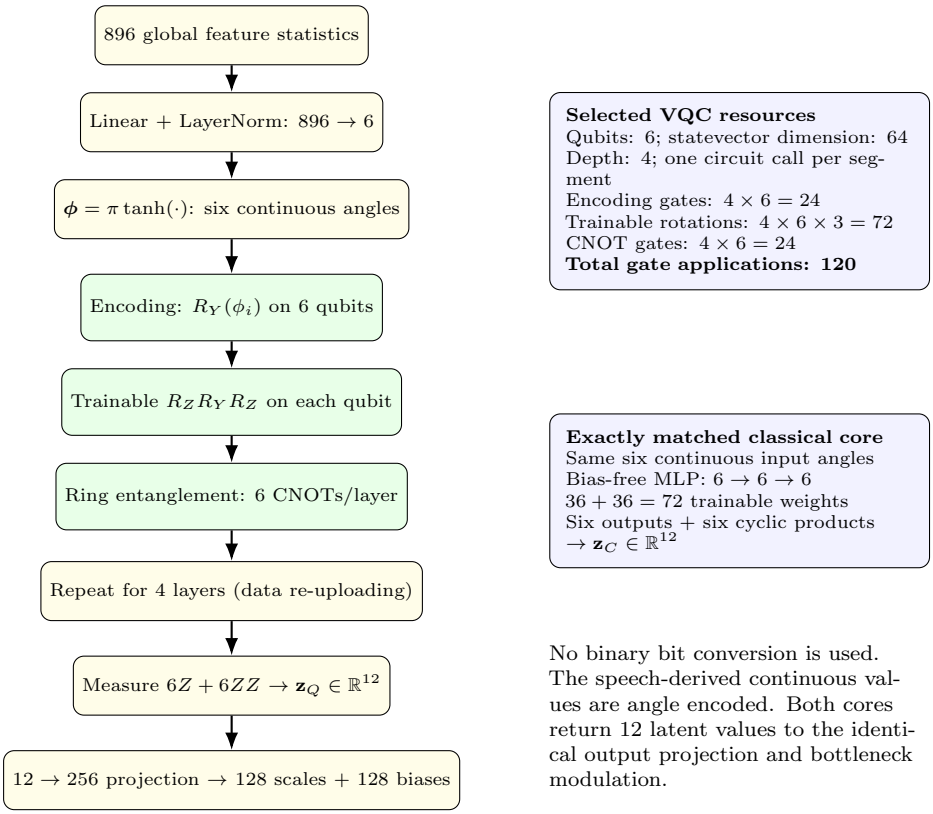


Figure 2: Quantum encoding and matched classical control. Six continuous bottleneck features are angle encoded on six qubits, processed by a depth-four variational circuit with ring entanglement and data re-uploading, and measured through six Z and six nearest-neighbor ZZ observables.

1.3 Classical Versus Quantum Conditioner

The selected QNN uses six qubits, circuit depth four, ring entanglement, data re-uploading, and one circuit call per speech segment. The initial state is $|000000\rangle$. At each of the four layers, the six projected features are angle encoded by one $R_Y(\phi_i)$ gate per qubit. Each qubit then undergoes a trainable Euler-style rotation sequence

$$R_Z(\theta_{l,i,1})R_Y(\theta_{l,i,2})R_Z(\theta_{l,i,3}), \tag{4}$$

followed by a six-edge ring of CNOT gates. The resource count is therefore 24 encoding R_Y gates, 72 trainable single-qubit rotation gates, and 24 CNOT gates, totaling 120 gate applications per circuit evaluation.

After the circuit, six single-qubit Z_i expectation values and six nearest-neighbor Z_iZ_{i+1} expectation values are evaluated,

$$\mathbf{z}_Q = [\langle Z_1\rangle, \ldots, \langle Z_6\rangle, \langle Z_1Z_2\rangle, \ldots, \langle Z_6Z_1\rangle] \in \mathbb{R}^{12}. \tag{5}$$

These 12 values are continuous latent features returned to the neural network. A common output projection maps them to 256 values, which are interpreted as 128 channel-wise scales and 128 channel-wise biases. The conditioned bottleneck is

$$\widetilde{H} = H \odot (1 + \alpha\,\mathbf{a}) + \alpha\,\mathbf{b}, \tag{6}$$

where $\mathbf{a}$ and $\mathbf{b}$ are bounded by $\tanh(\cdot)$ and $\alpha = 0.25\,\sigma(\gamma)$ is a learned conditioning strength.

The classical conditioned model receives the same six values $\boldsymbol{\phi}$ but replaces the quantum circuit with a bias-free two-layer MLP,

$$\mathbf{h} = \tanh(W_2 \tanh(W_1\boldsymbol{\phi})), \tag{7}$$

with dimensions $6 \to 6 \to 6$. Since each weight matrix contains $6 \times 6 = 36$ parameters, the classical core also has exactly 72 trainable parameters. To match the 12 outputs of the QNN, the six MLP outputs are concatenated with their cyclic neighboring products,

$$\mathbf{z}_C = [h_1, \ldots, h_6, h_1h_2, h_2h_3, \ldots, h_6h_1] \in \mathbb{R}^{12}. \tag{8}$$

Thus, the QNN and classical models have identical input projections, output projections, conditioning location, conditioner parameter counts, and total trainable parameter counts; only the internal 72-parameter transformation differs.

Table 2: Architectural and parameter comparison.

Property	Baseline	Classical	QNN
Total trainable parameters	521,028	529,823	529,823
Conditioner parameters	0	8,795	8,795
Core parameters	0	72	72
Conditioning location	None	Bottleneck	Bottleneck
Core input / output	–	$6 \to 12$	$6 \to 12$
Qubits	–	–	6
Circuit depth	–	–	4
Circuit calls / segment	–	–	1
Statevector dimension	–	–	64
Encoding gates	–	–	24
Trainable rotation gates	–	–	72
CNOT gates	–	–	24
Measured observables	–	–	$6Z + 6ZZ$

1.4 Locked-Test Results

Table 3 reports the final locked-test means and standard deviations across the three seeds. Both conditioned variants clearly outperform the unconditioned baseline. The classical conditioner already provides a substantial gain over the baseline, while the QNN achieves the best mean value for all three main objective metrics.

Table 3: Final locked-test performance averaged across the three seeds (mean ± standard deviation across seed-level means). Best values are shown in bold.

Model	Enhanced SI-SDR (dB) ↑	PESQ ↑	ESTOI ↑
Baseline	14.9815 ± 0.0997	1.9447 ± 0.0163	0.72680 ± 0.00069
Classical	15.8237 ± 0.0774	2.0969 ± 0.0188	0.74276 ± 0.00107
QNN	$\mathbf{15.8649 \pm 0.0926}$	$\mathbf{2.1133 \pm 0.0055}$	$\mathbf{0.74320 \pm 0.00086}$

In terms of improvement over the noisy input, the baseline achieves ΔSI-SDR = 7.4690 dB, ΔPESQ = 0.5090, and ΔESTOI = 0.10003. These increase to 8.3112 dB, 0.6612, and 0.11599 for the classical model and to 8.3523 dB, 0.6776, and 0.11643 for the QNN. Therefore, the locked-test mean QNN–classical differences are

$$\Delta_{Q-C} = +0.0411 \text{ dB (SI-SDR)}, \quad +0.01640 \text{ (PESQ)}, \quad +0.000434 \text{ (ESTOI)}. \tag{9}$$

The paired statistical analysis shows that the PESQ improvement is highly consistent across utterances, with 95% bootstrap confidence interval $[0.01486, 0.01805]$ and Holm-corrected one-sided $p = 1.74 \times 10^{-98}$. The ESTOI difference is also significant, with confidence interval $[0.000278, 0.000582]$ and corrected $p = 6.31 \times 10^{-4}$. SI-SDR has a positive mean confidence interval, $[0.0295, 0.0528]$ dB, but the paired Wilcoxon-Holm test is not significant ($p = 0.179$), indicating that the SI-SDR benefit is positive in mean but not uniformly distributed across utterances.

1.5 SNR-Wise Performance

Table 4 reports the locked-test enhanced SI-SDR, PESQ, and ESTOI for each SNR level. The same trend is visible at every SNR value: the classical model improves clearly over the baseline, and the QNN further improves over the classical model. The QNN has the highest mean value at all six SNRs for all three reported metrics.

Table 4: SNR-wise locked-test performance averaged across the three seeds. For each metric and SNR, the best value is shown in bold.

SNR (dB)	Enhanced SI-SDR (dB) ↑			PESQ ↑			ESTOI ↑		
	Baseline	Classical	QNN	Baseline	Classical	QNN	Baseline	Classical	QNN
-5	6.1854	7.2700	**7.3131**	1.3109	1.4035	**1.4118**	0.50856	0.53484	**0.53530**
0	10.2074	11.1642	**11.2121**	1.5217	1.6569	**1.6721**	0.64208	0.66480	**0.66523**
5	13.4070	14.3140	**14.3806**	1.7743	1.9255	**1.9473**	0.72521	0.74175	**0.74225**
10	16.7198	17.4913	**17.5246**	2.0236	2.1971	**2.2168**	0.77860	0.79176	**0.79196**
15	19.9198	20.6490	**20.6752**	2.3817	2.5646	**2.5834**	0.83738	0.84665	**0.84713**
20	23.4435	24.0510	**24.0801**	2.6656	2.8418	**2.8559**	0.86595	0.87383	**0.87437**

The SNR-wise trend further supports the main conclusion. At every SNR, the QNN exceeds the classical model in mean SI-SDR, PESQ, and ESTOI. The strongest single-SNR gains are observed at 5 dB, where the QNN–classical differences are +0.0665 dB in SI-SDR, +0.0219 in PESQ, and +0.00050 in ESTOI. When the low-SNR subset (−5, 0, and 5 dB) is considered jointly, the QNN advantage is statistically significant for all three metrics: +0.0526 dB in SI-SDR ($p = 0.00653$), +0.01519 in PESQ ($p = 2.71 \times 10^{-62}$), and +0.000469 in ESTOI ($p = 0.00229$).

1.6 Computational Trade-off

The QNN improvement is a performance result rather than a computational-speed advantage. The updated warm-up-controlled latency analysis shows that, for short inputs, the QNN is slower than both the baseline and the matched classical conditioner. However, the key observation is that the QNN introduces an approximately constant additive overhead, so its *relative* latency penalty decreases strongly as the audio duration increases. Table 5 summarizes the long-duration latency analysis from 15 s to 180 s.

Table 5: Warm-up-controlled inference latency and real-time factor for long audio durations. Mean latency is reported over repeated runs on the evaluated GPU.

Audio duration	Baseline (ms)	Classical (ms)	QNN (ms)	QNN / Classical	QNN RTF
15 s	32.530	32.515	62.098	1.91×	0.004140
30 s	68.662	69.401	99.450	1.43×	0.003315
60 s	185.486	186.338	216.615	1.16×	0.003610
90 s	302.828	304.242	333.810	1.10×	0.003709
120 s	416.398	417.454	447.899	1.07×	0.003732
180 s	634.791	636.496	666.260	1.05×	0.003701

These results show a very clear trend. The absolute QNN overhead relative to the classical model remains close to 30 ms across all tested durations, whereas the shared U-Net backbone cost grows with input length. Consequently, the QNN/classical latency ratio drops from approximately 1.91× at 15 s to only 1.05× at 180 s. In other words, the QNN remains slower in absolute latency, but its relative overhead becomes small for long inputs. All three models remain comfortably faster than real time. Even for 180 s audio, the QNN real-time factor is only 0.003701, meaning that 3 minutes of speech are processed in about 0.666 s.

Peak GPU memory is also essentially identical across the classical and QNN variants for the same duration. At 180 s, both conditioned models require approximately 3.70 GB of GPU memory. Therefore, the computational penalty of the QNN arises primarily from statevector circuit evaluation time rather than from increased parameter count or memory footprint.

Overall, the results establish a strict three-way comparison: the baseline quantifies the performance of the shared U-Net without global conditioning; the classical model quantifies the gain obtained from a 72-parameter nonlinear conditioner; and the QNN replaces only this 72-parameter core with a six-qubit depth-four variational circuit. The

QNN achieves the best mean SI-SDR, PESQ, and ESTOI on the locked test and at every individual SNR level, with the clearest and most statistically consistent advantage appearing in PESQ and in the low-SNR regime. The evidence therefore supports a small but measurable task-level benefit of the variational quantum conditioner under exact parameter matching, while not implying a computational quantum advantage.